

# AERIS Expert Review Packet

## Anomaly

|                     |                               |
|---------------------|-------------------------------|
| Source / metric     | tceq / co                     |
| Observed / expected | 0.4 / 0.105405                |
| Time                | 2026-07-30 19:00 UTC          |
| Location            | 29.6697, -95.1285             |
| Severity            | severe                        |
| Detectors           | zscore, stl, isolation_forest |

## Stored evidence summary

| Source      | Metric             | Unit    | Entities / points | Range; mean                      | Nearest to event                        |
|-------------|--------------------|---------|-------------------|----------------------------------|-----------------------------------------|
| asos        | humidity           | percent | 7 / 712           | 39 to 100; mean 71.9373          | 49.74 at 2026-07-30 18:55Z; dt 5 min    |
| asos        | temperature        | degC    | 7 / 712           | 23.3333 to 36.1111; mean 29.7995 | 34 at 2026-07-30 18:55Z; dt 5 min       |
| asos        | wind_direction     | degree  | 7 / 875           | 0 to 280; mean 166.114           | 140 at 2026-07-30 18:55Z; dt 5 min      |
| asos        | wind_speed         | m/s     | 7 / 925           | 0 to 8.2311; mean 2.8742         | 2.05778 at 2026-07-30 18:55Z; dt 5 min  |
| noaa_gfs    | gh_500             | m       | 8 / 96            | 5909.24 to 5954.66; mean 5936.32 | 5940.05 at 2026-07-30 18:00Z; dt 60 min |
| noaa_gfs    | pbl_height         | m       | 8 / 96            | 48.9097 to 2294.53; mean 829.728 | 1709.36 at 2026-07-30 18:00Z; dt 60 min |
| noaa_gfs    | precipitable_water | mm      | 8 / 96            | 35.2236 to 49.1773; mean 41.2854 | 41.3172 at 2026-07-30 18:00Z; dt 60 min |
| noaa_gfs    | surface_pressure   | hPa     | 8 / 96            | 1007.06 to 1015.13; mean 1011.47 | 1012.71 at 2026-07-30 18:00Z; dt 60 min |
| noaa_gfs    | t_850              | degC    | 8 / 96            | 18.5842 to 22.3746; mean 20.7409 | 19.0563 at 2026-07-30 18:00Z; dt 60 min |
| noaa_gfs    | u_10m              | m/s     | 8 / 96            | -3.54486 to 2.31858; mean 0.527  | 1.08085 at 2026-07-30 18:00Z; dt 60 min |
| noaa_gfs    | v_10m              | m/s     | 8 / 96            | 0.680634 to 6.59256; mean 3.1879 | 2.54289 at 2026-07-30 18:00Z; dt 60 min |
| openaq      | ozone              | ppm     | 17 / 1151         | -0.004 to 0.059; mean 0.0183     | 0.033 at 2026-07-30 19:00Z; dt 0 min    |
| openaq      | pm10               | ug/m3   | 3 / 214           | 12 to 142; mean 36.7383          | 22 at 2026-07-30 19:00Z; dt 0 min       |
| openaq      | pm25               | ug/m3   | 66 / 2781         | 0 to 37.5; mean 6.1971           | 8.2 at 2026-07-30 19:00Z; dt 0 min      |
| openweather | cloud_cover        | percent | 5 / 360           | 0 to 100; mean 57.6444           | 100 at 2026-07-30 18:58Z; dt 1.2 min    |
| openweather | humidity           | percent | 5 / 360           | 43 to 94; mean 73.8472           | 56 at 2026-07-30 18:58Z; dt 1.2 min     |
| openweather | precipitation      | mm/h    | 5 / 360           | 0 to 0; mean 0                   | 0 at 2026-07-30 18:58Z; dt 1.2 min      |
| openweather | pressure           | hPa     | 5 / 360           | 1010 to 1016; mean 1013.34       | 1014 at 2026-07-30 18:58Z; dt 1.2 min   |
| openweather | temperature        | degC    | 5 / 360           | 25.16 to 37.3; mean 30.6402      | 35.73 at 2026-07-30 18:58Z; dt 1.2 min  |
| openweather | wind_direction     | degree  | 5 / 360           | 0 to 321; mean 188.417           | 235 at 2026-07-30 18:58Z; dt 1.2 min    |
| openweather | wind_speed         | m/s     | 5 / 360           | 0.45 to 6.55; mean 2.2363        | 1.79 at 2026-07-30 18:58Z; dt 1.2 min   |
| purpleair   | pm25               | ug/m3   | 64 / 4514         | 0 to 232; mean 8.1319            | 1.1 at 2026-07-30 18:59Z; dt 0 min      |

| Source     | Metric                       | Unit    | Entities / points | Range; mean                           | Nearest to event                              |
|------------|------------------------------|---------|-------------------|---------------------------------------|-----------------------------------------------|
| sentinel5p | s5p_aer_ai_granule_available | count   | 8 / 8             | 1 to 1; mean 1                        | 1 at 2026-07-30 19:36Z; dt 36.2 min           |
| sentinel5p | s5p_ch4_granule_available    | count   | 8 / 8             | 1 to 1; mean 1                        | 1 at 2026-07-30 19:36Z; dt 36.2 min           |
| sentinel5p | s5p_co_column                | mol/m^2 | 8 / 8             | 0.0256055 to 0.0286852; mean 0.0273   | 0.027437 at 2026-07-30 19:36Z; dt 36.2 min    |
| sentinel5p | s5p_co_granule_available     | count   | 8 / 8             | 1 to 1; mean 1                        | 1 at 2026-07-30 19:36Z; dt 36.2 min           |
| sentinel5p | s5p_hcho_column              | mol/m^2 | 7 / 7             | 0.000177906 to 0.0002266; mean 0.0002 | 0.000213446 at 2026-07-30 20:04Z; dt 64.6 min |
| sentinel5p | s5p_hcho_granule_available   | count   | 7 / 7             | 1 to 1; mean 1                        | 1 at 2026-07-30 20:04Z; dt 64.6 min           |
| sentinel5p | s5p_no2_column               | mol/m^2 | 4 / 4             | 3.65242e-05 to 4.98019e-05; mean 0    | 3.93606e-05 at 2026-07-30 20:04Z; dt 64.6 min |
| sentinel5p | s5p_no2_granule_available    | count   | 4 / 4             | 1 to 1; mean 1                        | 1 at 2026-07-30 20:04Z; dt 64.6 min           |
| sentinel5p | s5p_o3_granule_available     | count   | 8 / 8             | 1 to 1; mean 1                        | 1 at 2026-07-30 19:36Z; dt 36.2 min           |
| sentinel5p | s5p_so2_column               | mol/m^2 | 7 / 7             | -7.46696e-06 to 0.000148654; mean 0   | 2.45829e-05 at 2026-07-30 20:04Z; dt 64.6 min |
| sentinel5p | s5p_so2_granule_available    | count   | 7 / 7             | 1 to 1; mean 1                        | 1 at 2026-07-30 20:04Z; dt 64.6 min           |
| tceq       | co                           | ppm     | 3 / 191           | 0.1 to 0.8; mean 0.2126               | 0.4 at 2026-07-30 19:00Z; dt 0 min            |
| tceq       | no2                          | ppb     | 12 / 737          | -2.7 to 22.7; mean 4.2236             | 1.3 at 2026-07-30 19:00Z; dt 0 min            |
| tceq       | so2                          | ppb     | 3 / 192           | -0.7 to 5.7; mean -0.0026             | 0.1 at 2026-07-30 19:00Z; dt 0 min            |

## Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

| Source   | Metric             | Values                                                                                                                                                                 |
|----------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| asos     | humidity           | 07-29 06Z 89.48, 12Z 70.8, 18Z 51.51, 07-30 00Z 78.88, 06Z 90.59, 12Z 69.87, 18Z 50.89, 07-31 00Z 77.11, 06Z 89.38, 12Z 70.39, 18Z 47.41, 08-01 00Z 76.09, 06Z 88.89   |
| asos     | temperature        | 07-29 06Z 25.47, 12Z 30.4, 18Z 34.32, 07-30 00Z 28.85, 06Z 26.21, 12Z 30.39, 18Z 34.28, 07-31 00Z 28.61, 06Z 25.76, 12Z 30.05, 18Z 34.63, 08-01 00Z 28.66, 06Z 26.16   |
| asos     | wind_direction     | 07-29 06Z 81.86, 12Z 196.7, 18Z 192.3, 07-30 00Z 158.3, 06Z 109.9, 12Z 176, 18Z 180.6, 07-31 00Z 166.6, 06Z 134, 12Z 210.1, 18Z 184.2, 08-01 00Z 189.6, 06Z 154.5      |
| asos     | wind_speed         | 07-29 06Z 0.9064, 12Z 3.073, 18Z 4.234, 07-30 00Z 3.416, 06Z 1.444, 12Z 2.305, 18Z 3.827, 07-31 00Z 3.291, 06Z 1.583, 12Z 3.212, 18Z 3.263, 08-01 00Z 3.708, 06Z 2.101 |
| noaa_gfs | gh_500             | 07-29 12Z 5911, 18Z 5928, 07-30 00Z 5924, 06Z 5933, 12Z 5924, 18Z 5940, 07-31 00Z 5940, 06Z 5950, 12Z 5940, 18Z 5954, 08-01 00Z 5948, 06Z 5945                         |
| noaa_gfs | pbl_height         | 07-29 12Z 226.8, 18Z 2098, 07-30 00Z 1044, 06Z 368.2, 12Z 216.4, 18Z 1656, 07-31 00Z 772.6, 06Z 387.4, 12Z 257.7, 18Z 1656, 08-01 00Z 851.4, 06Z 421.9                 |
| noaa_gfs | precipitable_water | 07-29 12Z 39.15, 18Z 43.83, 07-30 00Z 46.42, 06Z 36.88, 12Z 38.47, 18Z 42.19, 07-31 00Z 44.89, 06Z 38.11, 12Z 39.58, 18Z 42.29, 08-01 00Z 44.5, 06Z 39.11              |
| noaa_gfs | surface_pressure   | 07-29 12Z 1010, 18Z 1011, 07-30 00Z 1009, 06Z 1011, 12Z 1012, 18Z 1013, 07-31 00Z 1010, 06Z 1013, 12Z 1014, 18Z 1013, 08-01 00Z 1011, 06Z 1012                         |
| noaa_gfs | t_850              | 07-29 12Z 20.58, 18Z 19.04, 07-30 00Z 20.47, 06Z 21.57, 12Z 21.14, 18Z 19.2, 07-31 00Z 21.06, 06Z 21.82, 12Z 21.48, 18Z 19.26, 08-01 00Z 21.18, 06Z 22.1               |
| noaa_gfs | u_10m              | 07-29 12Z 1.022, 18Z 1.273, 07-30 00Z -0.3884, 06Z 0.9249, 12Z 1.167, 18Z 0.3059, 07-31 00Z -1.585, 06Z 0.8129, 12Z 1.364, 18Z 1.239, 08-01 00Z -1.531, 06Z 1.719      |

| Source      | Metric                       | Values                                                                                                                                                                                           |
|-------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| noaa_gfs    | v_10m                        | 07-29 12Z 1.856, 18Z 2.96, 07-30 00Z 5.022, 06Z 3.22, 12Z 1.306, 18Z 2.605, 07-31 00Z 5.251, 06Z 3.272, 12Z 1.956, 18Z 2.554, 08-01 00Z 4.985, 06Z 3.268                                         |
| openaq      | ozone                        | 07-29 06Z 0.008412, 12Z 0.01348, 18Z 0.02853, 07-30 00Z 0.01801, 06Z 0.01094, 12Z 0.01402, 18Z 0.03224, 07-31 00Z 0.01692, 06Z 0.01064, 12Z 0.01481, 18Z 0.03329, 08-01 00Z 0.01677, 06Z 0.01317 |
| openaq      | pm10                         | 07-29 06Z 37.67, 12Z 58.22, 18Z 45.39, 07-30 00Z 26.22, 06Z 22.17, 12Z 51.61, 18Z 37.73, 07-31 00Z 26.78, 06Z 22.39, 12Z 48.72, 18Z 37.61, 08-01 00Z 32.12, 06Z 18.83                            |
| openaq      | pm25                         | 07-29 06Z 5.515, 12Z 6.945, 18Z 6.734, 07-30 00Z 7.054, 06Z 7.508, 12Z 7.395, 18Z 5.085, 07-31 00Z 5.42, 06Z 6.227, 12Z 7.092, 18Z 5.811, 08-01 00Z 3.994, 06Z 2.4                               |
| openweather | cloud_cover                  | 07-29 06Z 0.68, 12Z 13.83, 18Z 81.55, 07-30 00Z 52.9, 06Z 85.37, 12Z 72.52, 18Z 79.32, 07-31 00Z 85.33, 06Z 89.3, 12Z 86, 18Z 36.53, 08-01 00Z 5.633, 06Z 2.4                                    |
| openweather | humidity                     | 07-29 06Z 89.2, 12Z 78.34, 18Z 58.29, 07-30 00Z 74.53, 06Z 88.77, 12Z 78.69, 18Z 56.52, 07-31 00Z 73.87, 06Z 87.7, 12Z 77.4, 18Z 52.7, 08-01 00Z 72.7, 06Z 82.4                                  |
| openweather | precipitation                | 07-29 06Z 0, 12Z 0, 18Z 0, 07-30 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-31 00Z 0, 06Z 0, 12Z 0, 18Z 0, 08-01 00Z 0, 06Z 0                                                                                |
| openweather | pressure                     | 07-29 06Z 1012, 12Z 1013, 18Z 1011, 07-30 00Z 1012, 06Z 1013, 12Z 1015, 18Z 1013, 07-31 00Z 1014, 06Z 1015, 12Z 1016, 18Z 1014, 08-01 00Z 1013, 06Z 1014                                         |
| openweather | temperature                  | 07-29 06Z 25.99, 12Z 29.85, 18Z 35.49, 07-30 00Z 30.36, 06Z 26.84, 12Z 29.83, 18Z 35.77, 07-31 00Z 30.19, 06Z 26.54, 12Z 29.74, 18Z 36.1, 08-01 00Z 30.33, 06Z 27.52                             |
| openweather | wind_direction               | 07-29 06Z 164.4, 12Z 217.1, 18Z 192.4, 07-30 00Z 175.4, 06Z 186.9, 12Z 233, 18Z 188.2, 07-31 00Z 175.3, 06Z 158, 12Z 202, 18Z 182.6, 08-01 00Z 190, 06Z 154                                      |
| openweather | wind_speed                   | 07-29 06Z 1.662, 12Z 2.068, 18Z 2.284, 07-30 00Z 2.733, 06Z 2.038, 12Z 1.981, 18Z 2.143, 07-31 00Z 2.578, 06Z 2.035, 12Z 2.612, 18Z 2.057, 08-01 00Z 2.506, 06Z 2.43                             |
| purpleair   | pm25                         | 07-29 06Z 7.694, 12Z 8.734, 18Z 8.437, 07-30 00Z 9.693, 06Z 10.01, 12Z 8.863, 18Z 6.459, 07-31 00Z 7.379, 06Z 8.368, 12Z 9.307, 18Z 6.748, 08-01 00Z 6.326, 06Z 4.575                            |
| sentinel5p  | s5p_aer_ai_granule_available | 07-29 18Z 1, 07-30 18Z 1, 07-31 18Z 1                                                                                                                                                            |
| sentinel5p  | s5p_ch4_granule_available    | 07-29 18Z 1, 07-30 18Z 1, 07-31 18Z 1                                                                                                                                                            |
| sentinel5p  | s5p_co_column                | 07-29 18Z 0.02655, 07-30 18Z 0.02743, 07-31 18Z 0.02868                                                                                                                                          |
| sentinel5p  | s5p_co_granule_available     | 07-29 18Z 1, 07-30 18Z 1, 07-31 18Z 1                                                                                                                                                            |
| sentinel5p  | s5p_hcho_column              | 07-29 18Z 0.0002022, 07-30 18Z 0.0002134, 07-31 18Z 0.000219                                                                                                                                     |
| sentinel5p  | s5p_hcho_granule_available   | 07-29 18Z 1, 07-30 18Z 1, 07-31 18Z 1                                                                                                                                                            |
| sentinel5p  | s5p_no2_column               | 07-29 18Z 3.912e-05, 07-30 18Z 3.936e-05, 07-31 18Z 4.98e-05                                                                                                                                     |
| sentinel5p  | s5p_no2_granule_available    | 07-29 18Z 1, 07-30 18Z 1, 07-31 18Z 1                                                                                                                                                            |
| sentinel5p  | s5p_o3_granule_available     | 07-29 18Z 1, 07-30 18Z 1, 07-31 18Z 1                                                                                                                                                            |
| sentinel5p  | s5p_so2_column               | 07-29 18Z 6.457e-05, 07-30 18Z 2.458e-05, 07-31 18Z 2.122e-05                                                                                                                                    |
| sentinel5p  | s5p_so2_granule_available    | 07-29 18Z 1, 07-30 18Z 1, 07-31 18Z 1                                                                                                                                                            |
| tceq        | co                           | 07-29 06Z 0.2333, 12Z 0.2722, 18Z 0.1944, 07-30 00Z 0.21, 06Z 0.2, 12Z 0.2313, 18Z 0.1944, 07-31 00Z 0.1778, 06Z 0.2111, 12Z 0.2333, 18Z 0.2056, 08-01 00Z 0.1889, 06Z 0.1333                    |
| tceq        | no2                          | 07-29 06Z 6.305, 12Z 5.49, 18Z 3.479, 07-30 00Z 2.292, 06Z 3.799, 12Z 5.327, 18Z 3.942, 07-31 00Z 2.124, 06Z 4.48, 12Z 4.993, 18Z 4.299, 08-01 00Z 2.113, 06Z 3.583                              |
| tceq        | so2                          | 07-29 06Z -0.1867, 12Z -0.1722, 18Z -0.1333, 07-30 00Z -0.1222, 06Z -0.1889, 12Z 0.4278, 18Z -0.1556, 07-31 00Z -0.1444, 06Z -0.2, 12Z 0.3556, 18Z 0.4444, 08-01 00Z -0.08889, 06Z -0.2167       |

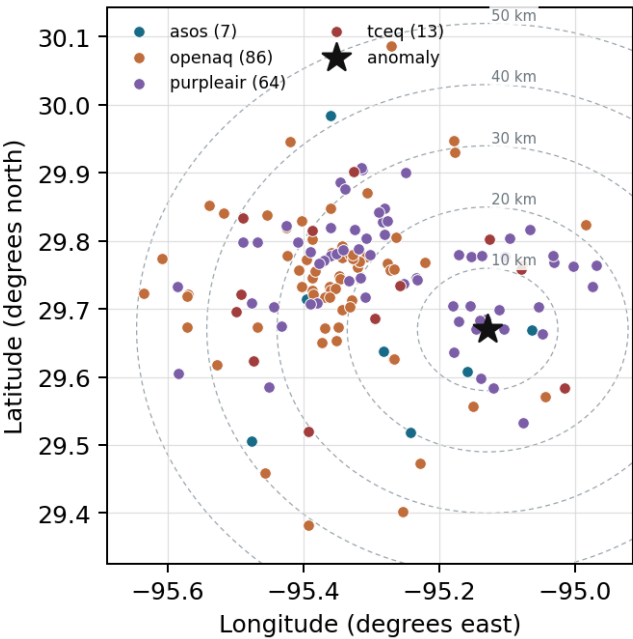
## Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

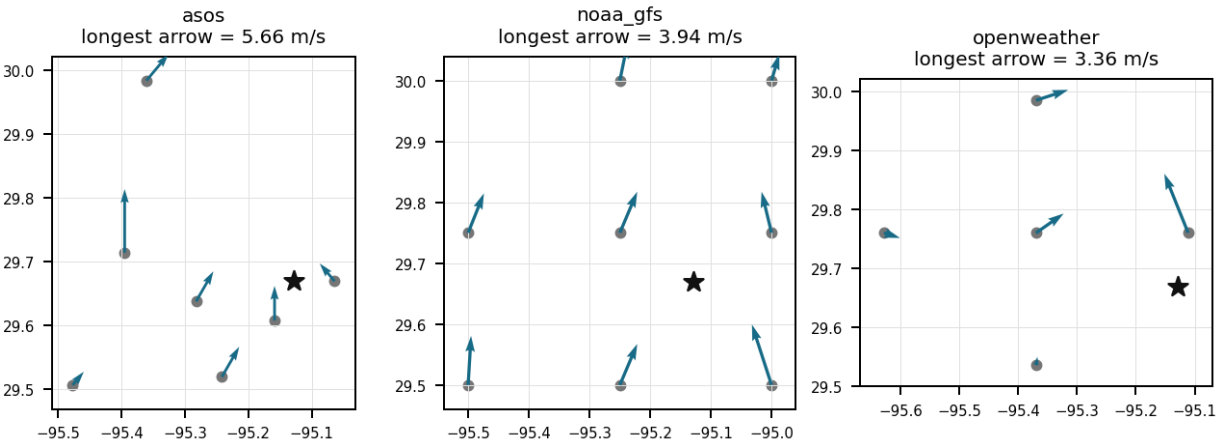
| Source      | Metric             | Values                                                                                                                                                                                                                                                                                              |
|-------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| asos        | humidity           | T41 (6.225 km) 73.78, EFD (7.53 km) 73.17, HOU (15.295 km) 72.23, LVJ (20.026 km) 72.23, MCJ (26.207 km) 70.68, AXH (38.286 km) 72.73, IAH (41.543 km) 67.56                                                                                                                                        |
| asos        | temperature        | T41 (6.225 km) 29.83, EFD (7.53 km) 29.91, HOU (15.295 km) 29.97, LVJ (20.026 km) 29.66, MCJ (26.207 km) 29.98, AXH (38.286 km) 29.05, IAH (41.543 km) 30.74                                                                                                                                        |
| asos        | wind_direction     | T41 (6.225 km) 197.3, EFD (7.53 km) 176.9, HOU (15.295 km) 185.1, LVJ (20.026 km) 137.4, MCJ (26.207 km) 185.9, AXH (38.286 km) 108, IAH (41.543 km) 185                                                                                                                                            |
| asos        | wind_speed         | T41 (6.225 km) 3.817, EFD (7.53 km) 3.698, HOU (15.295 km) 3.605, LVJ (20.026 km) 2.09, MCJ (26.207 km) 3.432, AXH (38.286 km) 1.194, IAH (41.543 km) 2.74                                                                                                                                          |
| noaa_gfs    | gh_500             | gfs:29.75,-95.25 (14.741 km) 5936, gfs:29.75,-95.00 (15.288 km) 5936, gfs:29.50,-95.25 (22.231 km) 5936, gfs:29.50,-95.00 (22.598 km) 5935, gfs:29.75,-95.50 (36.97 km) 5937, gfs:30.00,-95.25 (38.548 km) 5937, gfs:30.00,-95.00 (38.76 km) 5937, gfs:29.50,-95.50 (40.577 km) 5936                |
| noaa_gfs    | pbl_height         | gfs:29.75,-95.25 (14.741 km) 958.8, gfs:29.75,-95.00 (15.288 km) 863.3, gfs:29.50,-95.25 (22.231 km) 670.2, gfs:29.50,-95.00 (22.598 km) 711.5, gfs:29.75,-95.50 (36.97 km) 1073, gfs:30.00,-95.25 (38.548 km) 1002, gfs:30.00,-95.00 (38.76 km) 586.7, gfs:29.50,-95.50 (40.577 km) 771.6          |
| noaa_gfs    | precipitable_water | gfs:29.75,-95.25 (14.741 km) 41.2, gfs:29.75,-95.00 (15.288 km) 41.77, gfs:29.50,-95.25 (22.231 km) 41.25, gfs:29.50,-95.00 (22.598 km) 41.35, gfs:29.75,-95.50 (36.97 km) 40.69, gfs:30.00,-95.25 (38.548 km) 41.34, gfs:30.00,-95.00 (38.76 km) 41.84, gfs:29.50,-95.50 (40.577 km) 40.85         |
| noaa_gfs    | surface_pressure   | gfs:29.75,-95.25 (14.741 km) 1012, gfs:29.75,-95.00 (15.288 km) 1012, gfs:29.50,-95.25 (22.231 km) 1012, gfs:29.50,-95.00 (22.598 km) 1013, gfs:29.75,-95.50 (36.97 km) 1010, gfs:30.00,-95.25 (38.548 km) 1010, gfs:30.00,-95.00 (38.76 km) 1011, gfs:29.50,-95.50 (40.577 km) 1011                |
| noaa_gfs    | t_850              | gfs:29.75,-95.25 (14.741 km) 20.71, gfs:29.75,-95.00 (15.288 km) 20.68, gfs:29.50,-95.25 (22.231 km) 20.73, gfs:29.50,-95.00 (22.598 km) 20.88, gfs:29.75,-95.50 (36.97 km) 20.8, gfs:30.00,-95.25 (38.548 km) 20.8, gfs:30.00,-95.00 (38.76 km) 20.62, gfs:29.50,-95.50 (40.577 km) 20.7           |
| noaa_gfs    | u_10m              | gfs:29.75,-95.25 (14.741 km) 0.8419, gfs:29.75,-95.00 (15.288 km) 0.4702, gfs:29.50,-95.25 (22.231 km) 0.671, gfs:29.50,-95.00 (22.598 km) 0.7585, gfs:29.75,-95.50 (36.97 km) 0.8735, gfs:30.00,-95.25 (38.548 km) 0.1819, gfs:30.00,-95.00 (38.76 km) 0.1394, gfs:29.50,-95.50 (40.577 km) 0.2794 |
| noaa_gfs    | v_10m              | gfs:29.75,-95.25 (14.741 km) 3.365, gfs:29.75,-95.00 (15.288 km) 3.184, gfs:29.50,-95.25 (22.231 km) 3.123, gfs:29.50,-95.00 (22.598 km) 3.666, gfs:29.75,-95.50 (36.97 km) 3.221, gfs:30.00,-95.25 (38.548 km) 2.591, gfs:30.00,-95.00 (38.76 km) 2.871, gfs:29.50,-95.50 (40.577 km) 3.482        |
| openaq      | ozone              | 332 (0.032 km) 0.02071, 2800 (11.598 km) 0.01874, 2039 (14.096 km) 0.01875, 320 (14.272 km) 0.0154, 3378 (14.356 km) 0.01416, 5077791 (14.558 km) 0.01485, 3214 (14.765 km) 0.02151, 325 (16.163 km) 0.005079, +9 more                                                                              |
| openaq      | pm10               | 271 (0.032 km) 25.79, 13380082 (14.356 km) 62.45, 15852419 (21.358 km) 22.18                                                                                                                                                                                                                        |
| openaq      | pm25               | 268 (0.032 km) 9.923, 13955815 (12.727 km) 6.438, 14404954 (13.725 km) 4.472, 14157975 (14.275 km) 8.452, 3379 (14.356 km) 11.18, 15539682 (14.359 km) 8.479, 8967123 (14.413 km) 9.711, 8967479 (14.413 km) 9.247, +58 more                                                                        |
| openweather | cloud_cover        | grid:east (10.221 km) 58.11, grid:center (25.388 km) 58.6, grid:south (27.676 km) 54.17, grid:north (42.079 km) 59.81, grid:west (49.325 km) 57.54                                                                                                                                                  |
| openweather | humidity           | grid:east (10.221 km) 77.68, grid:center (25.388 km) 72.92, grid:south (27.676 km) 74.35, grid:north (42.079 km) 70.92, grid:west (49.325 km) 73.38                                                                                                                                                 |
| openweather | precipitation      | grid:east (10.221 km) 0, grid:center (25.388 km) 0, grid:south (27.676 km) 0, grid:north (42.079 km) 0, grid:west (49.325 km) 0                                                                                                                                                                     |
| openweather | pressure           | grid:east (10.221 km) 1014, grid:center (25.388 km) 1013, grid:south (27.676 km) 1013, grid:north (42.079 km) 1013, grid:west (49.325 km) 1013                                                                                                                                                      |
| openweather | temperature        | grid:east (10.221 km) 30.36, grid:center (25.388 km) 31.09, grid:south (27.676 km) 30.27, grid:north (42.079 km) 30.94, grid:west (49.325 km) 30.54                                                                                                                                                 |
| openweather | wind_direction     | grid:east (10.221 km) 196.2, grid:center (25.388 km) 201.8, grid:south (27.676 km) 198, grid:north (42.079 km) 157, grid:west (49.325 km) 189                                                                                                                                                       |
| openweather | wind_speed         | grid:east (10.221 km) 3.66, grid:center (25.388 km) 2.223, grid:south (27.676 km) 1.672, grid:north (42.079 km) 2.112, grid:west (49.325 km) 1.515                                                                                                                                                  |
| purpleair   | pm25               | 128007 (1.625 km) 2.23, 161689 (1.727 km) 7.523, 222589 (1.859 km) 8.143, 268165 (2.296 km) 8.875, 222593 (3.655 km) 7.892, 186263 (4.264 km) 8.935, 98633 (4.643 km) 3.474, 182191 (6.162 km) 8.566, +56 more                                                                                      |

| Source | Metric | Values                                                                                                                                                                                                                                       |
|--------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tceq   | co     | 482011039 (0.0 km) 0.1188, 482011035 (14.355 km) 0.1547, 482011052 (29.756 km) 0.3667                                                                                                                                                        |
| tceq   | no2    | 482011039 (0.0 km) 2.624, 482011015 (10.998 km) 4.489, 482011035 (14.355 km) 8.23, 482011050 (14.566 km) 0.5875, 482010026 (14.787 km) 5.602, 482010416 (16.162 km) 4.581, 482011052 (29.756 km) 12.24, 480391004 (30.447 km) 2.171, +4 more |
| tceq   | so2    | 482011039 (0.0 km) 0.1422, 482011035 (14.355 km) -0.08438, 482010051 (33.805 km) -0.06562                                                                                                                                                    |

Ground observation locations in the stored 72-hour context  
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



## Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

### Claim 1 of 36

Heavy vehicular traffic emissions along nearby transportation corridors, such as State Highway 225, contributed directly to the elevated carbon monoxide concentrations recorded at the surface.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 2 of 36

The air-quality pattern near 29.670, -95.129 at 2026-07-30 19:00 UTC is most consistent with a short-lived local combustion plume because nearby ozone was 0.033 ppm, nearby PM10 was 22 ug/m3, nearby PM2.5 was 8.2 ug/m3, and broader regional PM2.5 remained modest rather than showing a coincident widespread smoke or aerosol event.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 3 of 36

The PM2.5 levels are elevated at 1.1 ug/m3, which is higher than the mean of 8.1319 ug/m3.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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#### Claim 4 of 36

A local primary combustion plume is supported because the TCEQ CO monitor at 29.670, -95.129 measured 0.4 ppm at 2026-07-30T19:00:00+00:00 versus an expected 0.105 ppm, while co-located TCEQ NO2 was only 1.3 ppb, co-located TCEQ SO2 was 0.1 ppb, nearby OpenAQ PM2.5 was 8.2 ug/m3, nearby OpenAQ PM10 was 22 ug/m3, and nearby OpenAQ ozone was 0.033 ppm rather than showing a parallel spike in multiple pollutants.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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#### Claim 5 of 36

Light surface wind speeds under 4 m/s restricted horizontal dispersion, facilitating the localized buildup of carbon monoxide emissions near the surface.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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#### Claim 6 of 36

A carbon monoxide concentration of 0.4 ppm was recorded at the TCEQ monitoring station at coordinates (29.670, -95.129) on July 30, 2026, at 19:00 UTC.

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|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 7 of 36

Local meteorology could have allowed a narrow source plume to intersect the monitor at 19 UTC because surface conditions were hot and relatively dry with light-to-moderate winds near the site, with ASOS reporting about 34 C, 49.74 percent humidity, wind from 140 degrees at 2.06 m/s near the anomaly time, while GFS showed southerly low-level flow and a daytime planetary boundary layer near 1709 m that would favor advection of a discrete plume rather than stagnant nighttime accumulation.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 8 of 36

The air quality anomaly in Houston, Texas is not caused by changes in wind direction or speed.

|                   |                            |                            |                            |
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| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 9 of 36

A nearby fresh combustion source such as on-road traffic, diesel equipment, marine or rail activity, or an industrial combustion plume is the leading physical explanation for the sharp CO enhancement because the TCEQ CO monitor at the anomaly site reached 0.4 ppm while colocated NO<sub>2</sub> was only 1.3 ppb, SO<sub>2</sub> was 0.1 ppb, PM<sub>2.5</sub> was 8.2 ug/m<sup>3</sup>, PM<sub>10</sub> was 22 ug/m<sup>3</sup>, and ozone was 0.033 ppm, which is consistent with a localized CO-rich plume rather than a broad regional pollution episode.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 10 of 36**

Meteorological transport from a source to the south or southeast of the monitor is supported because the nearest ASOS station reported 140 degree wind direction and 2.06 m/s wind speed within 5 minutes of the anomaly, the ASOS 18Z mean wind direction was about 181 degrees, and GFS 10 m winds near the site were positive u and v at 18Z, indicating southerly to southeasterly flow capable of carrying a narrow plume across the monitor.

|                          |                            |                            |                            |
|--------------------------|----------------------------|----------------------------|----------------------------|
| <b>Mark exactly one:</b> | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|--------------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 11 of 36**

The air quality anomaly in Houston, Texas is not caused by high levels of CO, NO<sub>2</sub>, SO<sub>2</sub>, or CH<sub>4</sub>.

|                          |                            |                            |                            |
|--------------------------|----------------------------|----------------------------|----------------------------|
| <b>Mark exactly one:</b> | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|--------------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 12 of 36**

The localized peak in surface carbon monoxide of 0.4 ppm recorded by TCEQ monitor 482011039, contrasted with baseline total column atmospheric CO concentrations measured by Sentinel-5P, points to a primary ground-level emission source near the monitor.

|                          |                            |                            |                            |
|--------------------------|----------------------------|----------------------------|----------------------------|
| <b>Mark exactly one:</b> | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|--------------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 13 of 36**

The TCEQ monitor at 29.670, -95.129 measured a severe local CO spike of 0.4 ppm at 2026-07-30T19:00:00+00:00 against an expected value near 0.105 ppm, while the same site measured only 1.3 ppb NO2 and 0.1 ppb SO2 at that time.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 14 of 36**

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 15 of 36**

Low co-located nitrogen dioxide concentrations observed by TCEQ alongside modest fine particulate matter readings from OpenAQ and PurpleAir indicate that the carbon monoxide anomaly was not caused by a major regional combustion or wildfire event.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 16 of 36**

The carbon monoxide concentration at the monitor location reached 0.4 ppm compared to an expected baseline of approximately 0.105 ppm, corresponding to a z-score of 8.55.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 17 of 36**

Industrial combustion or process flaring at petrochemical facilities along the Houston Ship Channel corridor released a localized carbon monoxide plume, causing the TCEQ monitor to record 0.4 ppm at 19:00 UTC on July 30, 2026.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 18 of 36**

At the exact time of the carbon monoxide anomaly, adjacent air quality monitoring recorded an ozone concentration of 0.033 ppm and a PM2.5 concentration of 8.2 ug/m3.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 19 of 36**

The tceq carbon monoxide value of 0.4 ppm at 2026-07-30T19:00:00+00:00 at monitor 482011039 exceeded the mean carbon monoxide level of 0.1188 ppm for that monitor over the provided context window.

**Mark exactly one:**☐ V☐ I☐ U**Note:**

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**Claim 20 of 36**

The anomaly occurred on July 30th at around 18:58:49 UTC.

**Mark exactly one:**☐ V☐ I☐ U**Note:**

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**Claim 21 of 36**

Surface and boundary-layer meteorology near Houston at the anomaly time supported transport of a narrow plume from the south to southeast toward the monitor rather than regional accumulation, because nearby observed winds were about 140 degrees at 2.06 m/s and modeled daytime planetary boundary layer height was about 1709 m.

**Mark exactly one:**☐ V☐ I☐ U**Note:**

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### Claim 22 of 36

Regional stagnation or a large area combustion episode is contradicted because GFS planetary boundary layer height near the site was about 1709 m at 18Z, surface winds were not calm in ASOS and OpenWeather observations, Sentinel-5P CO column near the site was only 0.02744 mol/m<sup>2</sup> without an obvious jump relative to adjacent days, and particle measurements from OpenAQ and PurpleAir were not broadly elevated at the anomaly time.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 23 of 36

The NO<sub>2</sub> levels were also elevated in Houston, Texas on July 30th, with a peak value of 22.7 ppb at monitor 482011039.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 24 of 36

The air quality anomaly is related to PM<sub>2.5</sub> levels.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 25 of 36

A large regional smoke or urban-scale accumulation event is less likely as the primary cause of the CO spike because satellite CO column near Houston on 2026-07-30 was only modestly elevated relative to adjacent days, satellite SO<sub>2</sub> and NO<sub>2</sub> columns were not strongly enhanced, and local particle measurements remained low to moderate rather than showing the widespread PM increase expected from regional smoke.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 26 of 36

At 2026-07-30T19:00:00+00:00 at 29.670, -95.129, the nearest reported co-pollutant values were 1.3 ppb for tceq nitrogen dioxide, 0.1 ppb for tceq sulfur dioxide, 0.033 ppm for openaq ozone, 22.0 ug/m<sup>3</sup> for openaq PM<sub>10</sub>, and 8.2 ug/m<sup>3</sup> for openaq PM<sub>2.5</sub>.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 27 of 36

Surface weather observations from ASOS stations recorded light south-southeasterly winds under 4 m/s during the anomaly window, which restricted local atmospheric dispersion.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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Claim 28 of 36

The temperature anomaly is caused by a heatwave.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

Claim 29 of 36

The tceq carbon monoxide monitor at 29.670, -95.129 recorded 0.4 ppm at 2026-07-30T19:00:00+00:00, compared with an expected value of 0.1054054054054054 ppm, corresponding to a z-score of 8.550663492024825 and a severe anomaly classification.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

Claim 30 of 36

A severe carbon monoxide concentration anomaly was detected on July 30, 2026, at 19:00 UTC at location (29.670, -95.129).

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

Claim 31 of 36

The PM2.5 levels are elevated due to nearby industrial activities or wildfires.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

### Claim 32 of 36

The PM2.5 levels exceeded the normal range in Houston, Texas on July 30th, with a peak value of 1.625 ug/m3 at sensor 128007.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 33 of 36

Sentinel-5P total column CO measurements and surface PM2.5 sensor networks showed stable, low concentrations across the Houston area, confirming the localized nature of the CO anomaly.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 34 of 36

Local surface wind speeds were measured between 2.0 and 3.8 m/s while the boundary layer height reached approximately 1650 to 1700 meters during the afternoon of July 30, 2026.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
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Note:

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### Claim 35 of 36

The wind direction and speed anomalies are caused by a low-pressure system.

|                   |                            |                            |                            |
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| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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Claim 36 of 36

The air quality anomaly in Houston, Texas on July 30th was caused by a combination of factors including high temperatures, low wind speeds, and nearby industrial activities.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
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Note:

**Optional: most likely cause**

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.

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